

Homework 3

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Set up

Problem 1

a) An Appropriate Test

To determine the strength of the relationship between ocean temperature (°C) and giant kelp frond elongation rate (cm day⁻¹), appropriate tests would be a **Pearson correlation** and a **Spearman rank correlation**. Pearson correlation measures the strength and direction of a **linear relationship** between the two continuous variables, while Spearman rank correlation measures the strength and direction of a **monotonic relationship** based on ranked data and is less sensitive to non-normal data and outliers.

b) Visualization

```
ggplot(temp_kelp,
  aes(x = temp_c,
      y = kelp_elong)
) +

geom_point(
  color = "darkgreen",
  size = 1
) +

geom_smooth(
  method = "lm",
  se = TRUE,
  color = "steelblue"
) +
```

```
labs(
  x = "Ocean temperature (°C)",
  y = expression("Giant kelp elongation rate (cm day"^-1*")"),
  title = "Relationship between ocean temperature and giant kelp growth"
) +

theme_bw()
```

`geom\_smooth()` using formula = 'y ~ x'

